

ST3008 · APPLIED STATISTICAL MODELS

Laptop Price Prediction Using Regression Models

Analysis of laptop prices using Multiple Linear Regression and a Proportional Odds Model

Group 03

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Why Model Laptop Prices?

The Problem

Laptop prices vary considerably across brands and specifications, making it hard for consumers to judge value. Statistical modeling helps identify which hardware and software features drive price, and supports purchasing, pricing, and marketing decisions.

Dataset source: Kaggle(Laptop Price dataset)

<https://www.kaggle.com/datasets/muhammetvarl/laptop-price?resource=download>

Study Objectives

- 1 Perform descriptive analysis and visualize price distributions and relationships
- 2 Examine associations between laptop specifications and market price
- 3 Build a Multiple Linear Regression model and a Proportional Odds model to predict price
- 4 Evaluate fitted models using diagnostic methods and verify assumptions
- 5 Interpret significant predictors and identify the most influential factors

Dataset Overview

1,303

laptops in the final analysis

12

original variables recorded

19

manufacturers (Company)

Key Variables

Variable	Description	Type / Levels
Company	Laptop manufacturer (e.g. Dell, HP, Apple)	Categorical (19)
Product	Brand / model name — excluded (high cardinality)	Categorical (618)
TypeName	Notebook, Ultrabook, Gaming, Workstation, 2-in-1	Categorical (6)
ScreenResolution	Resolution, panel tech, touchscreen	Categorical (40)
Cpu / Gpu	Processor / graphics manufacturer & model	Categorical (118 / 110)
Ram, Weight, Inches	RAM (GB), weight (kg), screen size (in)	Continuous
Memory	Storage capacity and type (HDD/SSD/Hybrid/Flash)	Categorical (39)
OpSys	Operating system installed	Categorical (9)
Price	Market selling price (response variable)	Continuous

Feature Engineering

High-cardinality categorical variables were regrouped into broader, interpretable categories to reduce dimensionality.

Cpu_Group

118 levels grouped by text pattern into Intel Core i3/i5/i7/M, Xeon, Celeron, Pentium, Atom, AMD A-/E-Series, Ryzen, FX, Other

Gpu_Group

110 levels grouped into Intel Iris/Integrated, AMD Radeon/FirePro, Nvidia MX/GTX/Quadro/Other

Storage_Type & Total_Storage_GB

Memory field split into storage technology (SSD, HDD, Hybrid, Flash, SSD+HDD...) and total capacity in GB

Resolution_Group, Touchscreen, IPS

ScreenResolution split by X_res/Y_res into HD, Full HD, QHD/Retina, 4K, plus binary touchscreen & IPS flags

Ram_GB & Weight_kg

Unit strings ('GB', 'kg') stripped to create clean numeric variables

Company_Group & Price_Group

Rare brands grouped as 'Other'; Price_euros split into ordered Low / Medium / High tiers

Final Predictors for Modelling

Excluded from final models

- Product — 618 unique values; risk of overfitting
- OpSys_Group — "Mac" almost exactly overlaps with "Apple", redundant with Company_Group
- Total_Storage_GB, Touchscreen, IPS — no significant additional explanatory power after forward/backward selection

Response variables

Price_euros — continuous price, used in the MLR model

Price_Group — Low / Medium / High tiers (tertile split at the 33.3rd & 66.6th percentiles), used in the ordinal model

Final predictor set (identical for both models)

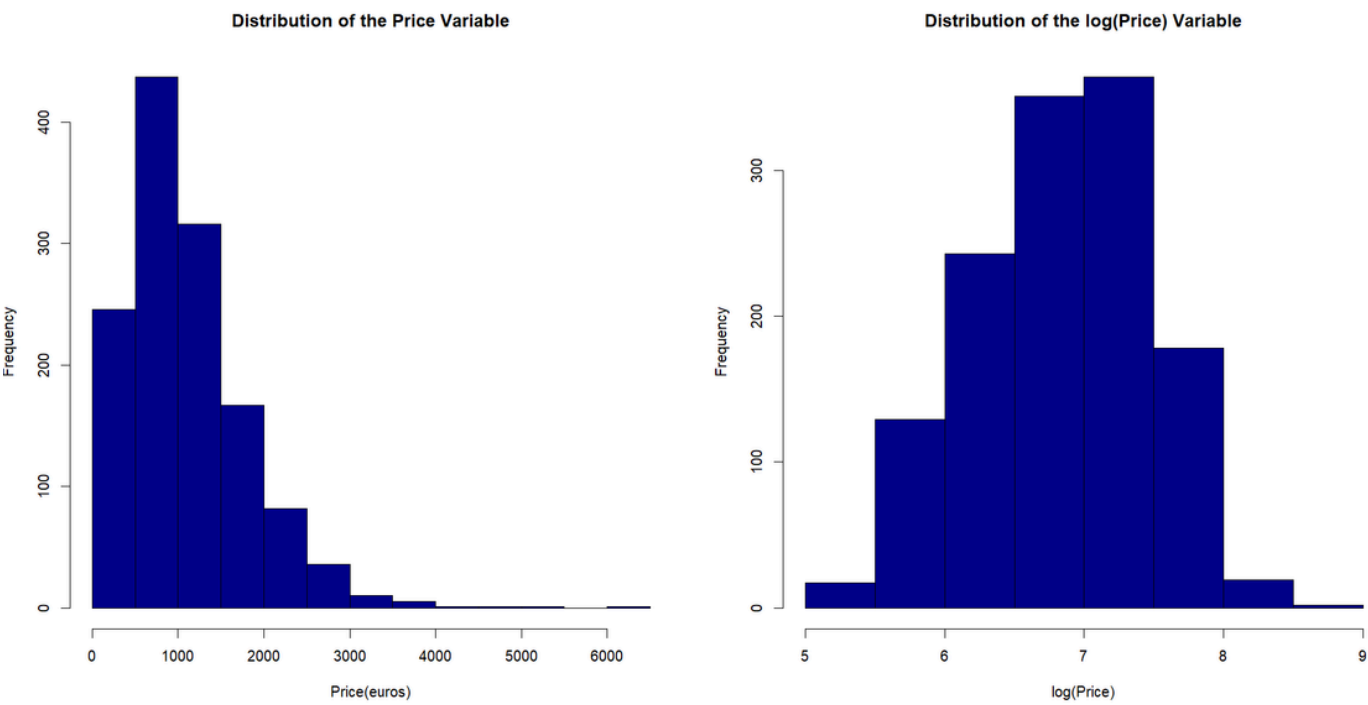
Company_Group	TypeName
Inches	Ram_GB
Weight_kg	Cpu_Group
Gpu_Group	Storage_Type
Resolution_Group	

Both forward selection (AIC) and backward elimination (F-test) converged on this same set of predictors for the MLR model.

Price Distributions & Relationships

Distributions

Price is right-skewed, driven by a few high-value laptops. The log transformation of price produces a more balanced, symmetric distribution. Weight and screen size show more concentrated patterns with less variation.



Illustrative right-skew of price / RAM / storage

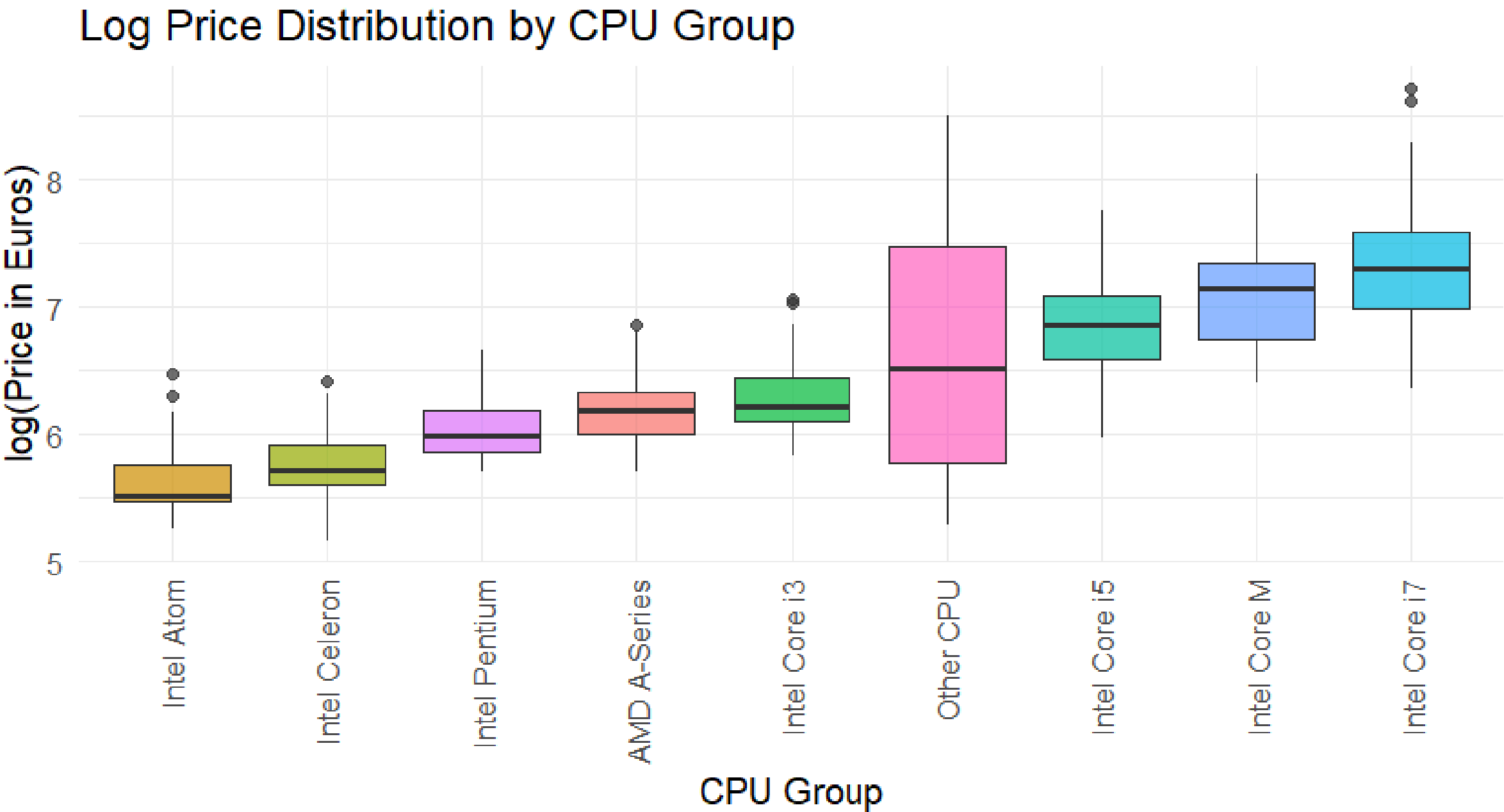
Relationships with Price

RAM	Strongest positive relationship — more RAM, higher price
Screen size	Slight positive trend with price
Weight	Weaker relationship, prices spread across values
Storage	Weaker relationship, prices spread across values

Log-Price by Category (Boxplots)

CPU Group

Intel Core i7 / i5 laptops have the highest median prices; Intel Atom / Celeron the lowest. "Other CPU" shows the widest spread.

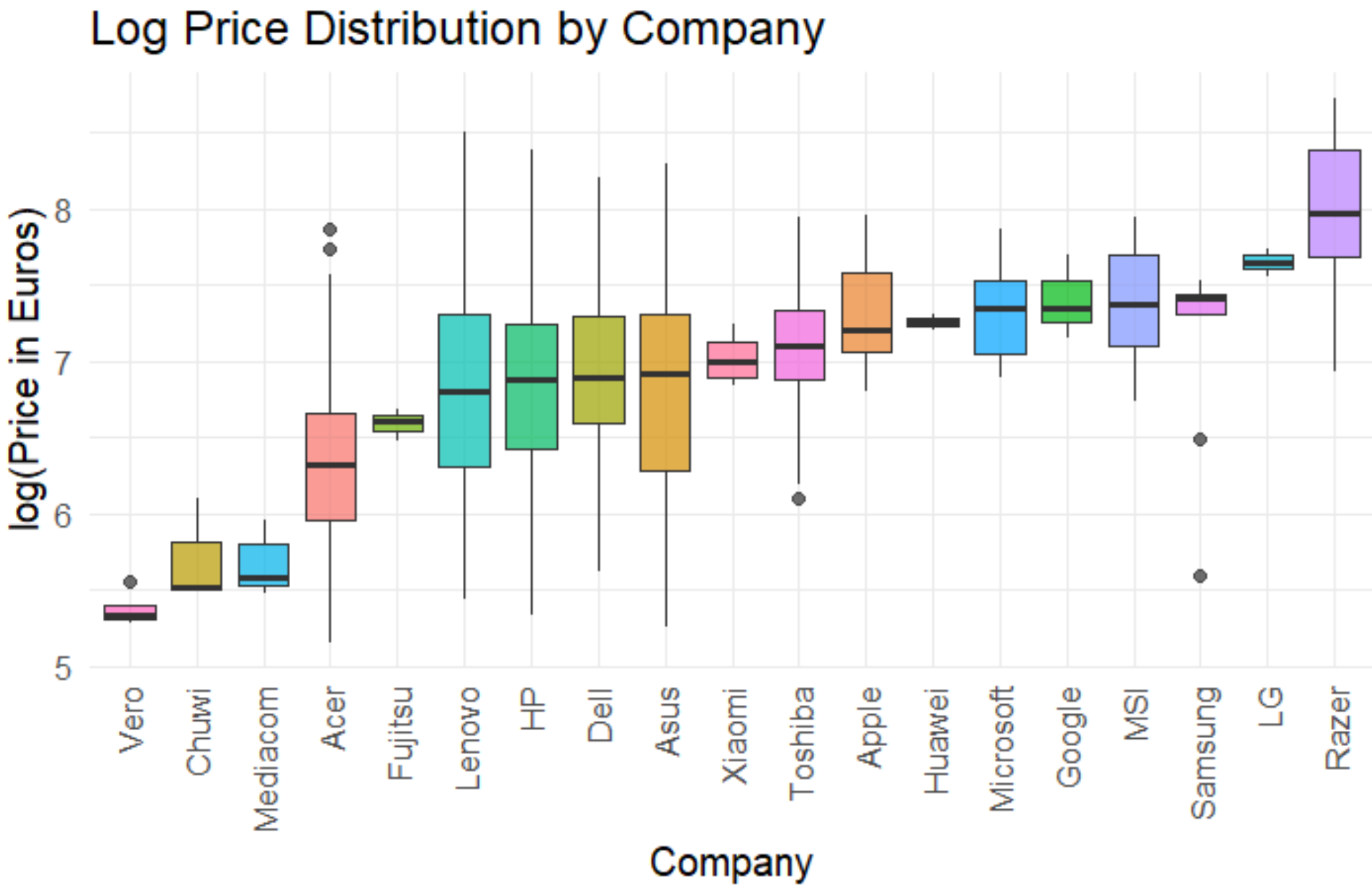


Outliers (unusually priced models) were observed within several CPU groups and for brands such as Acer and Samsung.

Log-Price by Category (Boxplots)

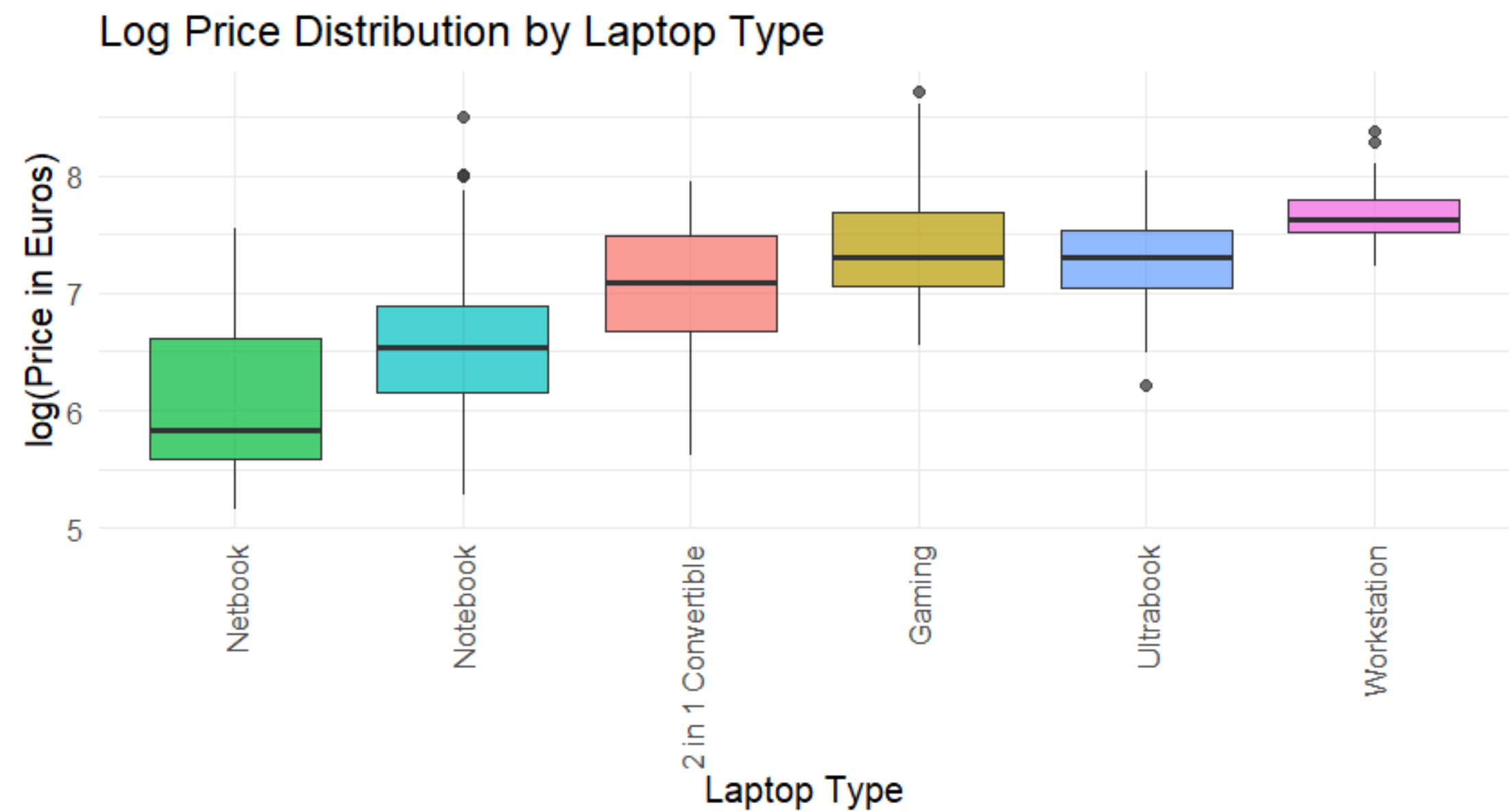
Company

Razer and LG have the highest median prices; Vero, Chuwi, and Mediacom the lowest. Acer and Dell span the widest price range.



Outliers (unusually priced models) were observed within several CPU groups and for brands such as Acer and Samsung.

Log-Price by Category (Boxplots)



Laptop Type

Workstations have the highest median price, followed by Gaming and Ultrabook. Netbooks are the most budget-friendly.

Outliers (unusually priced models) were observed within several CPU groups and for brands such as Acer and Samsung.

Building the MLR Model

Adjusted R²

Response: log(Price_euros)

Backward elimination (F-test) and forward selection (AIC) were both used to identify predictors, and both approaches converged on the same final model: Company_Group, TypeName, Inches, Ram_GB, Weight_kg, Cpu_Group, Gpu_Group, Storage_Type, and Resolution_Group.

The model explains 82.8% of the variation in log(Price_euros) (Adjusted R² = 82.8%). The intercept (6.7158, ≈ €826 on the original scale) is not practically interpretable on its own, since a laptop with zero inches, zero RAM, and zero weight is unrealistic — it mainly anchors the equation.

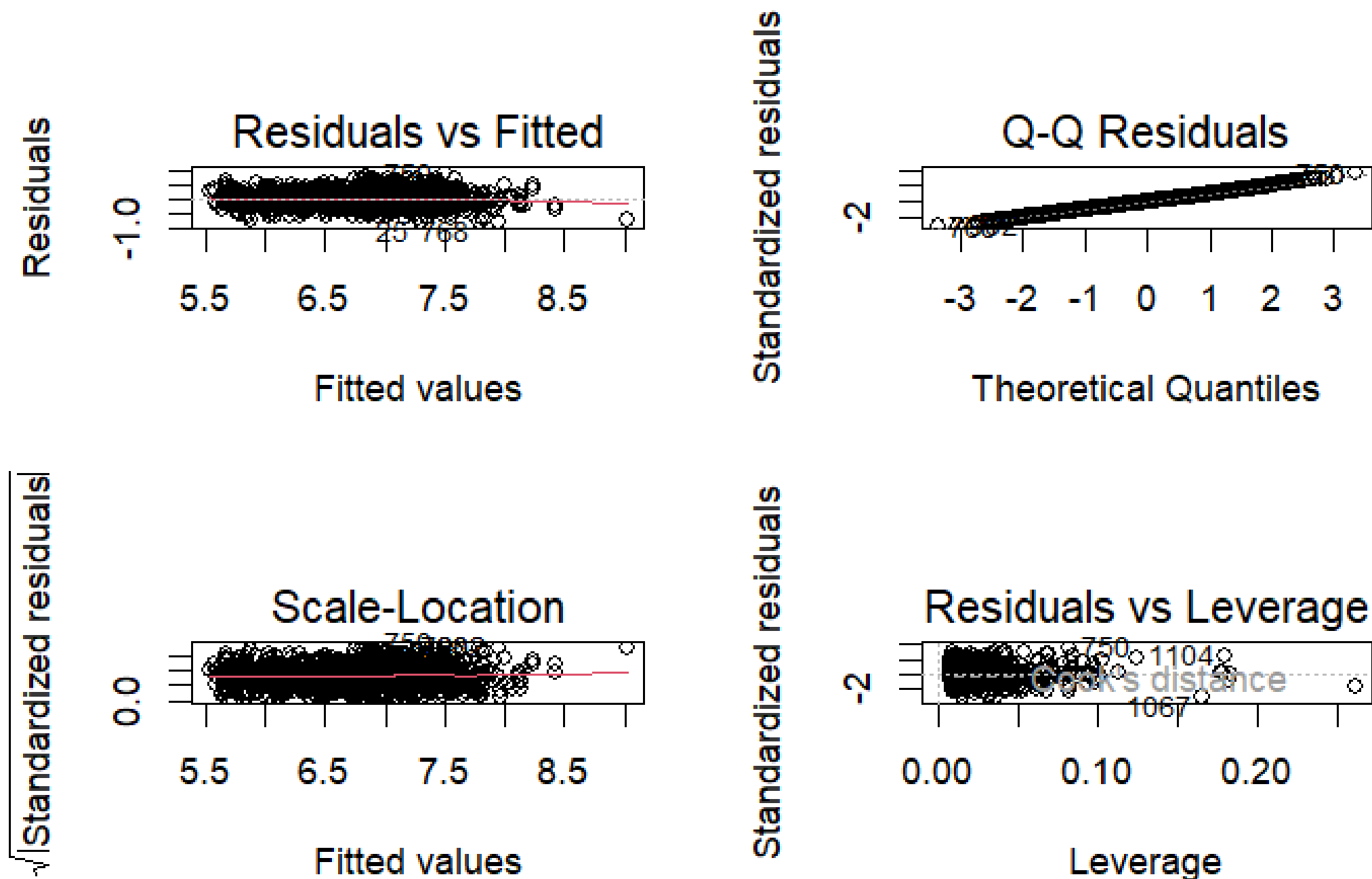
$$\begin{aligned} \hat{\text{Price}} = \exp(& \underbrace{6.7158}_{\text{Intercept}} + \beta(\underbrace{\text{Company}}_{\text{Group}}) + \beta(\underbrace{\text{TypeName}}_{\text{Type}}) - 0.0470 \cdot \underbrace{\text{Inches}}_{\text{Size}} + 0.0336 \cdot \underbrace{\text{Ram_GB}}_{\text{Memory}} + \\ & 0.0986 \cdot \underbrace{\text{Weight_kg}}_{\text{Weight}} + \beta(\underbrace{\text{Cpu}}_{\text{Group}}) + \beta(\underbrace{\text{Gpu}}_{\text{Group}}) + \beta(\underbrace{\text{Storage}}_{\text{Type}}) + \beta(\underbrace{\text{Resolution}}_{\text{Type}})) \end{aligned}$$

Key Effects on Price (log scale)

Percentage change in expected price, holding other predictors constant

Factor	Reference	Strongest effect
Company	Acer	Toshiba +27.3%, Apple +25.2% (largest premiums)
Laptop Type	2-in-1 Convertible	Gaming/Workstation & Ultrabook +10.0%; Notebook/Netbook −14.1%
Screen size (Inches)	—	+1 inch → −4.6% price
RAM (GB)	—	+1 GB → +3.4% price
Weight (kg)	—	+1 kg → +10.4% price
CPU	AMD	Intel Core i7 +49.8% (strongest); Intel Low-End −32.2%
GPU	AMD	Nvidia Quadro/Other +54.9% (strongest); Intel Integrated/Iris +18.5%
Storage Type	Flash Storage	SSD+HDD +39.6% (strongest); SSD +31.6%
Screen Resolution	4K Ultra HD	HD −32.3% (largest discount); Full HD −24.6%

Residual Diagnostics



Overall: linearity, normality, homoscedasticity, and absence of influential observations are reasonably satisfied — the MLR model is appropriate for predicting laptop prices.

From Continuous Price to Price Tiers

Creating Price_Group

Price_euros was split into three ordered tiers using a tertile split at the 33.3rd and 66.6th percentiles, so each tier — Low, Medium, High — contains a virtually equal number of observations, preserving the natural rank order Low < Medium < High.



Modelling approach

Forward selection produced the same predictor set as the MLR model. Since TypeName violates the proportional odds assumption, it is modelled as a nominal effect, giving a partial proportional odds model.

Cumulative Logit Model

$$\log(P(Y \leq j) / P(Y > j)) = \alpha_j - \eta, \quad j = 1, 2$$

η includes Ram_GB, Company_Group, Cpu_Group, Gpu_Group, Storage_Type, Resolution_Group, and Inches.

Threshold parameters (nominal TypeName effect)

$$\begin{aligned} \alpha_1^* &= 1.0572 - \overset{\wedge}{2.6450}(\text{Gaming/Workstation}) - \\ &\quad 0.1405(\overset{\wedge}{\text{Notebook/Netbook}}) - 1.8435(\overset{\wedge}{\text{Ultrabook}}) \\ \alpha_2^* &= 3.9498 - \overset{\wedge}{0.2951}(\text{Gaming/Workstation}) + \\ &\quad 1.1384(\overset{\wedge}{\text{Notebook/Netbook}}) - \overset{\wedge}{0.3293}(\text{Ultrabook}) \end{aligned}$$

Reference categories: Acer · AMD CPU · AMD GPU · Flash Storage · 4K Ultra HD

Odds of a Higher Price Category

Odds ratios for belonging to a higher price tier (Low < Medium < High)

Factor	Reference	Odds ratio highlights
RAM (GB)	—	OR ≈ 1.34 per additional GB (+34.5% odds)
Company	Acer	Toshiba OR ≈ 13.97, Apple OR ≈ 9.73
CPU	AMD	Intel Core i7 OR ≈ 49.76 (strongest); i3 & Low-End not significant
GPU	AMD	Quadro/Other OR ≈ 10.43; Integrated/Iris OR ≈ 4.09; MX not significant
Storage Type	Flash Storage	SSD+HDD OR ≈ 9.69; SSD OR ≈ 6.10; HDD & Hybrid not significant
Screen Resolution	4K Ultra HD	HD OR ≈ 0.048 (−95.2% odds); Full HD OR ≈ 0.125 (−87.5% odds)
Screen size (Inches)	—	OR ≈ 0.739 per inch (−26.1% odds)
Laptop Type (nominal)	2-in-1 Convertible	Gaming/Workstation & Ultrabook shift the Low↔Medium threshold; Notebook/Netbook shifts Medium↔High

Model Adequacy & Performance

77.97%

overall classification accuracy

0.2218

Mean Absolute Error

1313.4

AIC (partial PO model)

Classification by price tier (of 1,303 laptops)

Tier	Correctly classified	Accuracy
Low	370 of 437	84.7%
Medium	301 of 432	69.7%
High	345 of 434	79.5%

Significance tests

vs. null model: LR $\chi^2 = 1621.5$, df = 34, p < 2.2×10^{-16} — predictors significantly improve fit

Partial vs. standard PO model: LR = 10.664, df = 3, p = 0.0137 — nominal TypeName effect improves fit

AIC comparison: 1313.435 (partial PO) vs 1318.1 (standard PO) — partial PO model preferred

Nearly all misclassifications occur between adjacent tiers (Low–Medium or Medium–High), preserving the ordinal structure of price.

CONCLUSION

Two Complementary Views of Laptop Pricing

MLR — Continuous Price

Explains 82.8% of variation in $\log(\text{Price})$. Higher RAM, weight, premium CPU/GPU tiers, SSD-based storage, and premium brands raise price; larger screens and lower resolution lower it.

OLR — Price Category

Classifies laptops into Low/Medium/High with 77.97% accuracy. The partial proportional odds model (nominal TypeName) fits significantly better than the standard model.

Both models consistently identify RAM, processor, graphics card, storage configuration, display resolution, and manufacturer brand as the primary determinants of laptop price.

Limitations

- MLR shows mild heteroscedasticity; OLR required a partial (not full) proportional odds specification
- Some categorical levels had few observations, reducing precision of certain estimates
- Dataset lacks battery life, processor generation, release year, build quality, warranty, and demand data

Thank you — Group 03